BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: MRD-11TH 240623 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** MRD-11TH 240623 (A)
VERSION: C

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. Passage of solvent molecules from dilute into concentrated solution is:

(A) Fluidity
(B) Osmosis
(C) Diffusion
(D) Reverse osmosis

Q2. Change in concentrations of reactants and products with change in time is:

(A) Order of reaction
(B) Rate of reaction
(C) Molecularity of reaction
(D) Rate constant

Q3. Reactions which never go to completion and finally attain a state of equilibrium, are:

(A) Endothermic
(B) Exothermic
(C) Reversible
(D) Irreversible

Q4. Existence of different crystalline solids in the same crystal form/shape is:

(A) Allotropy
(B) Isotropy
(C) Polymorphism
(D) Isomorphism

Q5. Boiling point of a liquid is not affected by:

(A) External pressure
(B) Altitude
(C) Amount liquid
(D) Impurities

Q6. Which one of the following is more temperature sensitive:

(A) Ionic crystals
(B) Covalent Crystals
(C) Metallic crystals
(D) Liquid crystals

Q7. Evaporation of a liquid is a spontaneous process that causes

(A) Cooling
(B) Heating
(C) Constant temperature
(D) Constant pressure

Q8. One mole of NaCl contains 6.02×10^{23}:

(A) ions
(B) Atoms
(C) Molecules
(D) Formula Units

Q9. A yield of reaction that is independent of the reaction conditions, is:

(A) Actual
(B) Theoretical
(C) Percent
(D) Standard

Q10. Principal quantum number (n) describe which property of electron-cloud:

(A) Energy
(B) Shape
(C) Orientation
(D) Spinning

Q11. Carbon in carbon dioxide molecule is hybridized.

(A) sp^3
(B) sp^2
(C) sp
(D) dsp^2

Q12. The geometrical shape of a triatomic molecule with zero dipole moment is:

(A) Linear
(B) Angular
(C) Trigonal planar
(D) Tetrahedral

Q13. The compressibility factor (Z) for an ideal gas is always.

(A) Zero
(B) Equal to one
(C) Less than one
(D) Greater than one

Q14. One molal equeous solution of which compound has the lowest freezing point:

(A) Urea
(B) Glucose
(C) Sugar
(D) Sodium chloride

Q15. An increase in internal energy of a system may cause:

(A) Temperature rise
(B) Phase Change
(C) Chemical change
(D) All of these

Q16. Which of the following is calculated indirectly by Born Haber cycle.

(A) Internal energy
(B) Kinetic energy
(C) Lattice energy
(D) Activation energy

Q17. A substance that acts as oxidizing agent in the reaction $H_2+Cl_2 \rightarrow 2HCl$

(A) H_2
(B) Cl_2
(C) HCl
(D) None

Q18. The oxidation number of oxygen in super oxides is:

(A) $-1/2$
(B) -1
(C) -2
(D) $+2$

Section – B

Time Allowed: 2:40 Hours | **Maximum Marks:** 40

Q2. Write short answers to any ten (10) of the following parts. All parts carry equal marks:-

(i) Why actual yield of reaction is always lesser than its theoretical yield?

(ii) Calculate the number of molecules present in 51grams of NH_3 at S.T.P?

(iii) What is Planck's Quantum theory? What are its postulates?

(iv) What are the five series of spectral lines, observed for hydrogen atom?

(v) Define resonance? What are the basic conditions for writing resonating structures?

(vi) Define plasma. How it occurs in natures?

(vii) What is meant by Joule Thomson effect?

(viii) What are dipole-dipole forces?

(ix) Why crystalline solids are anisotropic in nature?

(x) How Kc value helps to predict the extent of reversible reaction?

(xi) Calculate the pH of 0.0005M $Ca(OH)_2$ solution.

(xii) Calculate the mole fraction of C_2H_5OH in one molal aqueous C_2H_5OH solution.

(xiii) Calculate the oxidation state of carbon in: (a) CH_4 (b) CH_3Cl (c) CH_2Cl_2 (d) $CHCl_3$

Section – C

Maximum Marks: 27

> Note: Answer any three (3) questions. All questions carry equal marks:-

Q3.

(i) What are limiting and excess reactants? How limiting reactant controls the quantity of products?

(ii) State and explain Moseley's law?

Q4.

(i) Explain the shapes of NH_3 and BF_3 on the basis of VSEPR theory.

(ii) Hydrogen gas diffuses through a porous plate at a rate of 500cm/min at 0^oC. What will be the rate of diffusion of oxygen through the same plate at 0^oC?

Q5.

(i) Why crystalline solids hare definite shapes and sharp melting points?

(ii) State and explain Le-Chatelier's principle.

Q6.

(i) State and explain Hess's law of constant heat summation.

(ii) Balance the following equations by ion-electron method: (a) $I^{-1} + NO_2 \rightarrow I_2 + NO$ (acidic medium) (b) $H_2O_2 + MnO_4^{-1} + H^{+1} \rightarrow Mn^{+2} + O_2 + H_2O$ (acidic medium)